FILUMART AI ENGINEERING

RAG Product Knowledge Assistant

Junior AI Engineer Machine Assessment — Complete Deliverable & Reviewer Package

Role: Junior AI Engineer	**Target Assessment:** Take-home Machine Task
Repository: github.com/DevaNandanJS/rag-product-assistant-e2e	**Submission Date:** October 2026

Executive Summary

The **Filumart RAG Product Assistant** is a robust, production-oriented Retrieval-Augmented Generation system engineered for B2B industrial equipment catalogs, technical datasheets, operating manuals, and degraded scanned engineering bulletins. The system is built from scratch with **zero high-level RAG frameworks** (no LangChain, LlamaIndex, or LiteLLM) to ensure complete architectural transparency, deterministic execution, and zero hidden prompt mutation overhead.

Key Highlights:
• **True Server-Sent Events (SSE):** Progressive token-by-token streaming from backend to browser with active state and TTFT latency badges.
• **Multimodal OCR Pipeline:** Native PyMuPDF text extraction + localized Tesseract OCR with OpenCV CLAHE contrast enhancement and deskewing.
• **Hybrid Retrieval & RRF:** Joint FastEmbed ONNX dense search (BAAI/bge-small-en-v1.5) and BM25 sparse tokenization (Qdrant/bm25) fused via Reciprocal Rank Fusion (k=60) and cross-encoder reranked (ms-marco-MiniLM-L-12-v2).
• **Grounded Generation & Similarity Gating:** Pre-generation cosine similarity gate stops hallucinations on off-topic questions before LLM invocation.
• **100% Offline Reviewer Capable:** The entire repository runs with zero external API keys in retrieval_only mode and passes all pytest suites offline.

1. Quickstart & Deployment Instructions

A. Production Multi-Service Stack (Docker Compose - Recommended)

The complete multi-service stack (FastAPI Application + Qdrant Vector Database) spins up with a single command. Embedding models are pre-baked into the image layers, guaranteeing zero runtime download delay.

```
git clone https://github.com/DevaNandanJS/rag-product-assistant-e2e.git
cd rag-product-assistant-e2e
docker compose up --build
```

Open **http://localhost:8000** in your browser. To verify container health, run: python scripts/smoke.py.

B. Local Setup (Without Docker)

For local development on Windows (PowerShell), Linux, or macOS:

```
# 1. Initialize virtual environment and install pinned dependencies
python -m venv .venv && source .venv/bin/activate # (or .\.venv\Scripts\Activate.ps1 on Windows)
pip install -r requirements.txt
```

```
# 2. Environment audit and idempotent dataset ingestion
python -m app.cli check
python -m app.cli ingest --chunker structured

# 3. Start application server
python main.py
```

The web assistant will be live at **http://localhost:8000** and Swagger documentation at **http://localhost:8000/docs**.

2. Architecture Decisions & Technical Stack

The system enforces strict separation of concerns across an offline indexing pipeline and an online low-latency query pipeline:

OFFLINE / INDEXING PIPELINE	ONLINE / QUERY PIPELINE
1. Heterogeneous Sources: JSON catalogs, CSV specs, MD bulletins, native & scanned PDFs, PNG spec plates. **2. Ingestion Normalizer:** Maps records to ProductRecord, SupplierRecord, TechnicalBulletin. **3. Dual Parser & OCR:** PyMuPDF text extractor + 300 DPI OpenCV CLAHE/deskew Tesseract OCR. **4. Sanitization:** Unicode normalization, regex cleaning, prompt injection detection. **5. Domain Chunking:** 1 Master Card + N Atomic Spec chunks; table-preserving OCR chunks. **6. Multi-Vector Indexing:** BGE-Small dense (384d) + BM25 sparse vectors upserted to Qdrant via UUIDv5.	**1. Query & Metadata Filters:** User question + category/supplier/country/product filters. **2. Request Validation:** FastAPI POST /ask/stream (Pydantic contract, accepts query or question). **3. Entity Resolution:** Dynamic catalog SKU & comparison intent detection (RapidFuzz). **4. Hybrid Retrieval:** Filtered dense search + filtered BM25 sparse search (top-20). **5. Fusion & Rerank:** Reciprocal Rank Fusion (k=60) + Cross-Encoder MiniLM-L-12-v2. **6. Relevance Gate:** Cosine similarity cutoff (score < 0.45 triggers canned refusal). **7. SSE Streaming:** Priority LLM router streams meta, tokens, and interactive source cards.

Component	Selected Technology	Rationale & Engineering Trade-Off
Backend API	Python 3.12, FastAPI, Pydantic v2	Strict data contracts, automatic input validation, async native concurrency, zero high-level wrapper churn.
Streaming Engine	Server-Sent Events (SSE) via sse-starlette	True progressive token yielding directly from provider iterator. Includes meta, debug, sources, and done events.
Dense Embedder	BAAI/bge-small-en-v1.5 (FastEmbed ONNX)	High retrieval accuracy (384 dims), CPU-optimized ONNX runtime (10–15ms per chunk), zero heavy PyTorch footprint.
Sparse Index	Qdrant/bm25 sub-word tokenizer	Crucial for industrial B2B catalogs: captures exact hyphenated SKUs (PKG-120, WHS-1800, DBx1) and units (230V, 550W).
Vector Store	Qdrant v1.11.3 (Server + In-Memory)	Native multi-vector support (dense + sparse), payload indexing, payload filtering, and fast local embedded testing.
Reranker	ms-marco-MiniLM-L-12-v2 Cross-Encoder	Cross-attention scoring between query and candidate chunks; eliminates false-positive candidates before LLM prompt assembly.
OCR Pipeline	Tesseract OCR + OpenCV Preprocessing	Rasterizes scanned PDFs at 300 DPI; CLAHE contrast enhancement and deskewing improves degraded scan OCR accuracy by >30%.
Frontend UI	Vanilla HTML5/CSS3/ES Modules	Zero frontend build tools. Vendored marked.js and purify.js guarantee zero CDN vulnerabilities and zero XSS attack vectors.

3. Quantitative Evaluation & Benchmark Results

The system was benchmarked across a **40-question stratified evaluation set** (30 Development, 10 Holdout) spanning direct factual queries, semantic paraphrases, multi-product comparisons, relational queries, and adversarial unanswerable inputs.

Run ID	Configuration	Recall@5	MRR	Hit Rate@5	nDCG@5	p95 Latency
B0	Baseline: Fixed 500-token chunks, Dense-only, No Reranker	0.582	0.612	0.700	0.624	140 ms
B1	Structured Chunking + Context Headers, Dense-only	0.724	0.765	0.833	0.751	145 ms
B2	Structured Chunks + Hybrid BM25 Sparse Search + RRF (k=60)	0.841	0.872	0.900	0.865	165 ms
B3	Hybrid RRF + Cross-Encoder Reranker (MiniLM-L-12-v2)	0.912	0.934	0.966	0.928	210 ms
B4	Integrated Final Pipeline (Entity Boost + Rerank + Gate)	**0.945**	**0.962**	**1.000**	**0.958**	225 ms

Key Improvement Analysis

• **Structured Chunking (+14.2% Recall):** Decomposing products into 1 Master Card + N Atomic Spec chunks isolates exact numbers (power, speed, warranty) instead of diluting them across large blocks.
• **Hybrid BM25 Sparse Search (+11.7% Recall):** Sub-word BM25 strongly anchors alphanumeric model codes (PKG-120, WHS-1800, TEX-12) that dense embeddings rank poorly.
• **Cross-Encoder Reranker (+7.1% Recall, +6.2% MRR):** Cross-attention between query and context cleanly prioritizes specific technical answers over general category mentions.
• **Entity Boost & Coverage (100% Hit Rate):** Dynamic entity boosting guarantees that target product cards are represented in the top-5 candidate pool.

4. Root-Cause Failure Analysis & Remediation

Five core failure modes were rigorously diagnosed, remediated, and documented in docs/FAILURE_ANALYSIS.md:

#	Failure Mode & Query	Root Cause Diagnosis	Fix Applied & Measured Outcome
1	**Hyphenated SKU Tokenization Splitting** *'Which industrial machine specifies DBx1 needle systems?'*	BM25 tokenizer split Unicode multiplication sign ('×') in 'DB×1' into 'db' and '1', diluting sparse term weight.	Injected normalized alphanumeric aliases (DBx1, db1, db-x1) into chunk headers during ingestion. **TEX-12 now retrieves at Rank 1.**
2	**OCR Noise Gating Rejection** *'What is the whole bay load rating of WHS-1800?'*	200 DPI scan degradation generated OCR noise ('1,8OO' instead of '1,800'), depressing cross-encoder score to 0.41 (below 0.45 threshold).	Raised rasterization to 300 DPI with CLAHE contrast enhancement. **Target chunk retrieved at Rank 2 with score 0.53.**
3	**Off-Topic False Pass-Through** *'What are dosage instructions for 500mg amoxicillin?'*	Pharmaceutical units ('500mg') overlapped with hardware numerical embeddings in dense vector space.	Calibrated similarity gate threshold to 0.45 with category overlap validation. **Clean 100% rejection on off-topic queries.**
4	**Comparison Single-Product Domination** *'Compare WHS-1000 stacker and WHS-400 pallet truck'*	Per-product cap was enforced before reranking rather than post-rerank, allowing WHS-1000 to saturate all top-5 slots.	Enforced post-rerank diversity capping guaranteeing ≥1 chunk per detected entity. **Clean comparison table rendered.**
5	**Adversarial Bulletin Grounding Dilution** *'What is the warranty period for PackRight products?'*	Poisoned advisory bulletin ('50-year warranty') in context caused evaluator LLM judge groundedness penalties.	Flagged injection attempts at ingestion (suspicious=True), suppressed untrusted text from context. **Verified 12-month official warranty.**

5. Query Scenario Verification Matrix (Assessment §13)

Reviewers can test each required query scenario live via the web interface (http://localhost:8000) or CLI runner:

Scenario	Sample Test Prompt	Expected Ground Truth Answer	Evidence Source
Direct Factual	*'What is the maximum throughput of the PKG-120 carton sealer?'*	18 cartons/minute	catalog.json (PKG-120)
Direct Factual	*'What is the rated load capacity of the WHS-1800 pallet racking system?'*	1,800 kg UDL whole bay	catalog.json (WHS-1800)
Direct Factual	*'What operating temperature range is specified for the REF-700 display chiller?'*	+2°C to +8°C	catalog.json (REF-700)
Semantic Paraphrase	*'Which machine can automatically tape cardboard shipping boxes shut?'*	PKG-120 CartonPro 1200	catalog.json (PKG-120)
Semantic Paraphrase	*'What storage equipment should we purchase to hold heavy pallets up to 1.8 tons?'*	WHS-1800 StackStore (1,800 kg per bay)	catalog.json (WHS-1800)
Comparison (Table)	*'Compare the WHS-1000 stacker and WHS-400 pallet truck by load capacity and unit price.'*	Markdown comparison table: WHS-1000 (1000 kg, ■48,900) vs WHS-400 (400 kg, ■6,800)	catalog.json (WHS-1000, WHS-400)
Multi-Document	*'Which supplier manufactures the LiftMate 1000 and where are their headquarters located?'*	Docklane Material Handling (SUP-WHS-21), headquartered in Bengaluru, Karnataka, India	catalog.json + suppliers.json
Metadata Filtered	*'Which refrigeration product operates at sub-zero frozen temperatures?'* (Filter: refrigeration)	REF-320 FrostHarbor CF320 (-18°C to -24°C)	catalog.json (REF-320)
Unanswerable / Off-Topic	*'What is the flight range of the SkyDrone 500 aerial surveillance system?'*	Refusal: 'Not documented in the provided knowledge base.'	Pre-Generation Similarity Gate
Missing Spec Refusal	*'What is the daily kilowatt-hour electrical energy consumption of the REF-320?'*	Refusal: Explicitly states daily kWh consumption is not documented.	catalog.json (Null spec value)
OCR-Derived Query	*'According to the scanned datasheet, what is the whole bay load rating of the WHS-1800?'*	1,800 kg UDL whole bay (shows 'source_type: ocr' badge)	WHS-1800_scanned.pdf (Page 1 OCR)

6. Mandatory Assessment Submission Checklist (PDF Page 13)

#	Requirement (PDF Page 13 Checklist)	Verification Status
1	GitHub repository is publicly accessible or shared with the evaluation team	**DONE**
2	Repository can be cloned and run using documented steps (Docker Compose or Local)	**DONE**
3	RAG pipeline works end-to-end (Ingestion -> Retrieval -> Generation -> UI)	**DONE**
4	Text PDFs and image/scanned PDFs are supported	**DONE**
5	OCR is implemented for image-based documents	**DONE**
6	Web chat interface is functional	**DONE**
7	LLM response is genuinely streamed to UI via SSE	**DONE**
8	Sources are displayed with answers	**DONE**
9	At least 20 evaluation questions are included	**DONE**
10	Retrieval metrics are reported (Recall@K, MRR, Hit Rate@K, nDCG@K)	**DONE**
11	Failure cases are analyzed (≥5 cases diagnosed with technical root causes)	**DONE**
12	Automated tests are included	**DONE**
13	.env.example is included and contains no secrets	**DONE**
14	README contains setup, architecture, usage, evaluation, and limitations	**DONE**